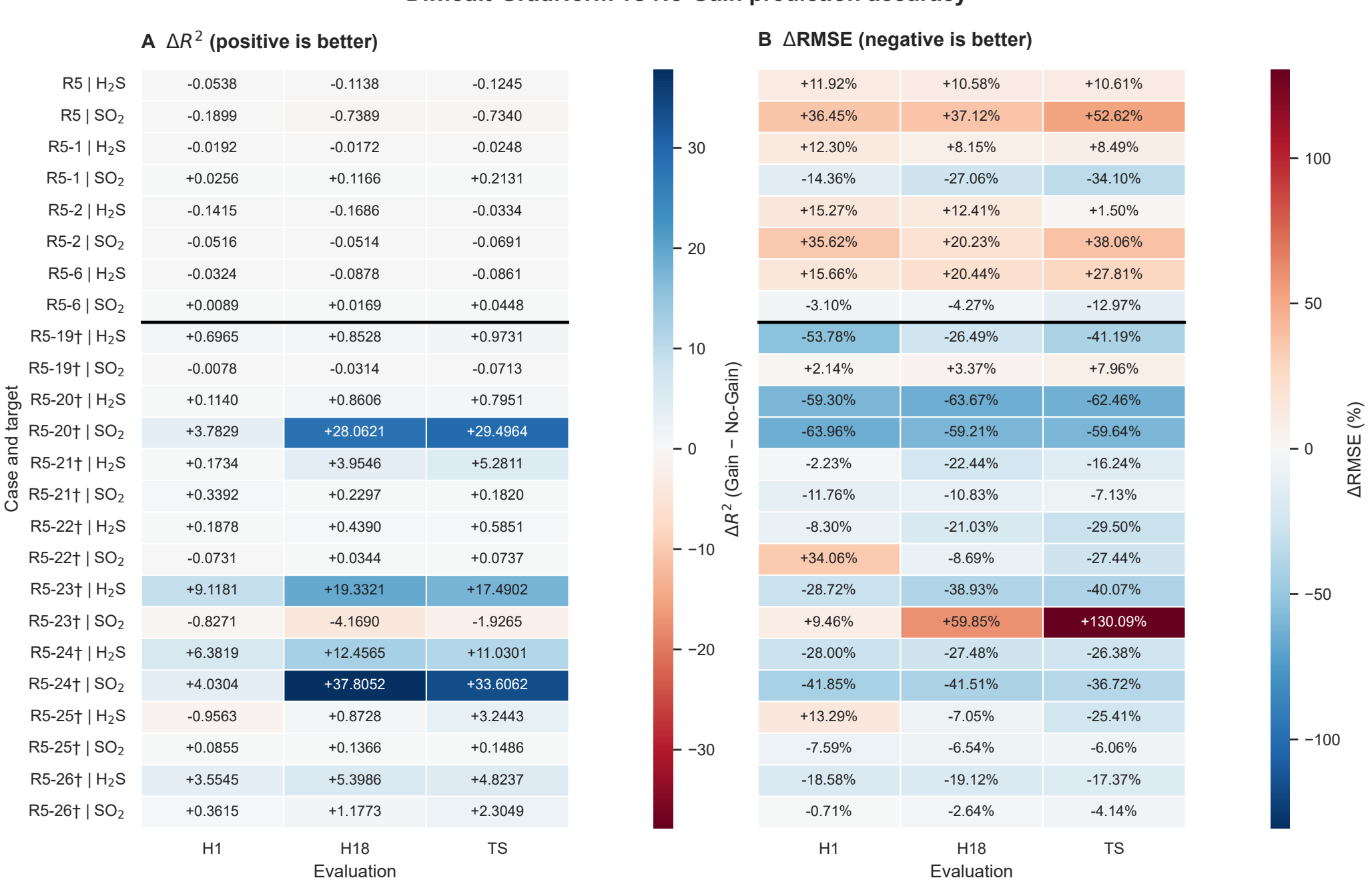

Difficult-GradNorm vs No-Gain prediction accuracy
A ΔR² (positive is better)
B ΔRMSE (negative is better)
Case and target
R5 | H2S
R5 | SO2
R5-1 | H2S
R5-1 | SO2
R5-2 | H2S
R5-2 | SO2
R5-6 | H2S
R5-6 | SO2
R5-19† | H2S
R5-19† | SO2
R5-20† | H2S
R5-20† | SO2
R5-21† | H2S
R5-21† | SO2
R5-22† | H2S
R5-22† | SO2
R5-23† | H2S
R5-23† | SO2
R5-24† | H2S
R5-24† | SO2
R5-25† | H2S
R5-25† | SO2
R5-26† | H2S
R5-26† | SO2
-0.0538 -0.1138 -0.1245
-0.1899 -0.7389 -0.7340
-0.0192 -0.0172 -0.0248
+0.0256 +0.1166 +0.2131
-0.1415 -0.1686 -0.0334
-0.0516 -0.0514 -0.0691
-0.0324 -0.0878 -0.0861
+0.0089 +0.0169 +0.0448
+0.6965 +0.8528 +0.9731
-0.0078 -0.0314 -0.0713
+0.1140 +0.8606 +0.7951
+3.7829 +28.0621 +29.4964
+0.1734 +3.9546 +5.2811
+0.3392 +0.2297 +0.1820
+0.1878 +0.4390 +0.5851
-0.0731 +0.0344 +0.0737
+9.1181 +19.3321 +17.4902
-0.8271 -4.1690 -1.9265
+6.3819 +12.4565 +11.0301
+4.0304 +37.8052 +33.6062
-0.9563 +0.8728 +3.2443
+0.0855 +0.1366 +0.1486
+3.5545 +5.3986 +4.8237
+0.3615 +1.1773 +2.3049
H1
H18
TS
Evaluation
ΔR² (Gain − No-Gain)
30
20
10
0
−10
−20
−30
+11.92% +10.58% +10.61%
+36.45% +37.12% +52.62%
+12.30% +8.15% +8.49%
-14.36% -27.06% -34.10%
+15.27% +12.41% +1.50%
+35.62% +20.23% +38.06%
+15.66% +20.44% +27.81%
-3.10% -4.27% -12.97%
-53.78% -26.49% -41.19%
+2.14% +3.37% +7.96%
-59.30% -63.67% -62.46%
-63.96% -59.21% -59.64%
-2.23% -22.44% -16.24%
-11.76% -10.83% -7.13%
-8.30% -21.03% -29.50%
+34.06% -8.69% -27.44%
-28.72% -38.93% -40.07%
+9.46% +59.85% +130.09%
-28.00% -27.48% -26.38%
-41.85% -41.51% -36.72%
+13.29% -7.05% -25.41%
-7.59% -6.54% -6.06%
-18.58% -19.12% -17.37%
-0.71% -2.64% -4.14%
H1
H18
TS
Evaluation
ΔRMSE (%)
100
50
0
−50
−100
Blue indicates improvement. † External all-data evaluation; the heatmaps show changes, not absolute prediction quality.